

Solar Site Suitability Mapping using Google Earth Engine

1. Introduction

Through this project I explored how remote sensing, geospatial analysis, and machine learning can be combined to identify suitable locations for solar energy deployment.

The work done in this project focuses on:

- Selecting an appropriate study region.
 - Integrating relevant geospatial datasets.
 - Framing the problem as a regression task.
 - Generating a continuous solar suitability map.
 - Validating the model using the maps.
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2. Study Region Selection (Week 2)

2.1 Rationale for Region Selection

To ensure sufficient spatial diversity and data availability, Gujarat was selected as the study region based on:

- High concentration of utility-scale solar power plants.
- Diverse terrain and land-cover conditions.
- Availability of consistent satellite and reanalysis data

2.2 Defining the Study Region

The study region was defined using administrative boundaries, specifically the state boundary of Gujarat. This ensured existing solar plant locations were mapped within the region to understand their spatial distribution. These locations were used for contextual understanding and validation, not as training labels.

3. Target Variable and Feature Selection (Week 3)

3.1 Target Variable

The dependent variable chosen for regression is **Global Horizontal Irradiance (GHI)**, which represents the total solar energy received on a horizontal surface. GHI was selected because it directly reflects solar energy availability and is widely used in solar resource assessment.

- Dataset: ERA5-Land Daily Aggregated.
- Band used: [surface_solar_radiation_downwards_sum](#).
- Temporal aggregation: Annual mean (2020)

3.2 Independent Variables

A minimal, physically meaningful set of predictor variables was used:

Feature	Source	Description
Elevation	SRTM DEM	Terrain height
Slope	Derived from DEM	Surface inclination
Aspect	Derived from DEM	Surface orientation
Land Cover	MODIS MCD12Q1	Surface type classification
Cloud proxy	ERA5-Land	Net solar radiation (atmospheric attenuation)
Surface Albedo	MODIS MCD43A3	Surface reflectivity

These features capture terrain, surface, and atmospheric influences on solar radiation.

4. Data Processing and Model Training

All datasets were clipped to the Gujarat boundary and resampled to a common spatial resolution (approximately 1 km). A pixel-wise sampling strategy was used to convert raster data into a training dataset, where each sample represents a location with associated physical features and corresponding GHI value.

Machine Learning Model

- Model type: Random Forest Regressor
- Output type: Continuous regression.

Random Forest was chosen due to its ability to:

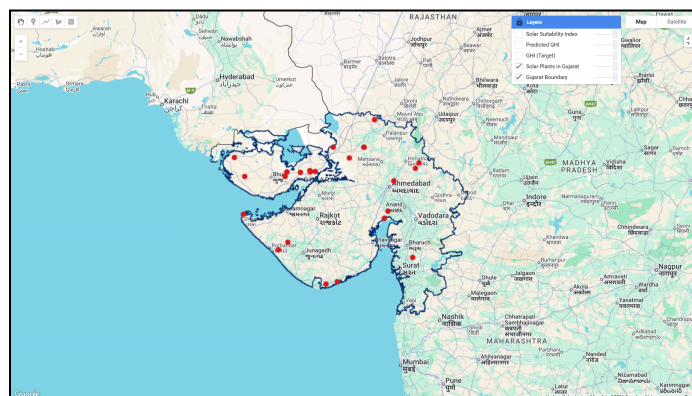
- Model nonlinear relationships.
 - Handle mixed data types.
 - Perform robustly with minimal hyperparameter tuning
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6. Model Output and Solar Suitability Mapping

The trained regression model was applied across the entire study region to generate a predicted GHI surface. To improve interpretability for decision-making, the predicted GHI values were normalized to create a Solar Suitability Index ranging from 0 to 1: 0 → Low suitability, 1 → High suitability. This continuous index allows flexible categorization into high, medium, and low suitability zones if required.

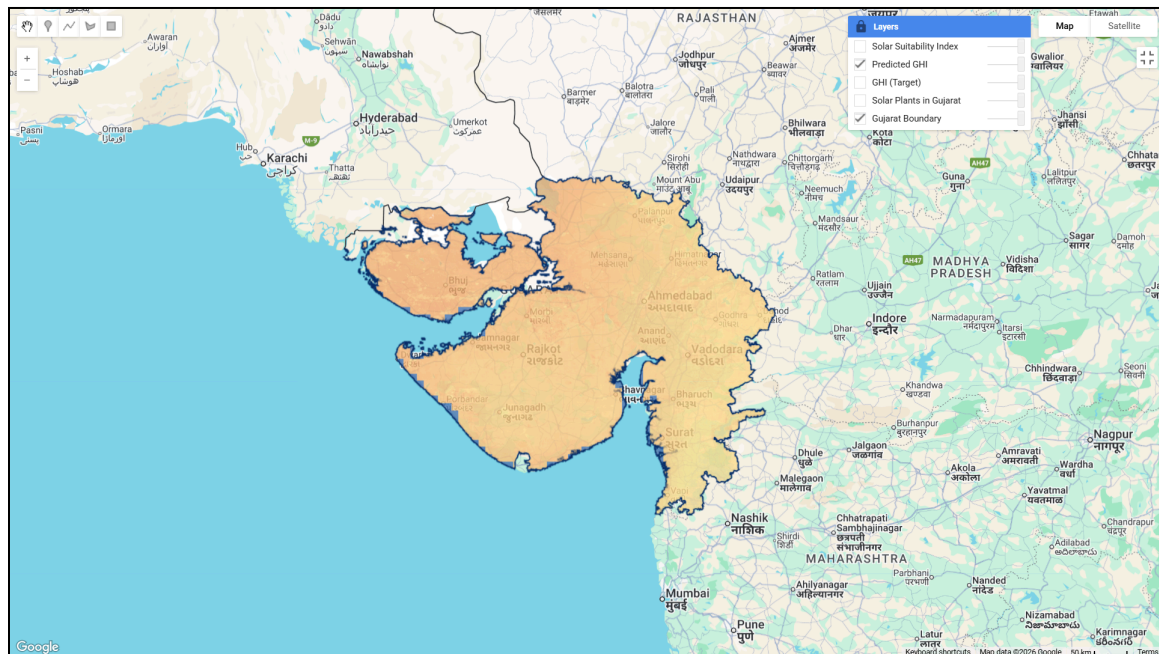
7. Resultant Maps

(I) Study Region & Solar plants Location



The blue line shows the boundary of Gujarat which is our defined study region. The red points show the existing solar plant locations in the defined study region.

(II) Predicted GHI map



This figure shows the Predicted Global Horizontal Irradiance (GHI) over Gujarat, as estimated by the Random Forest regression model trained using terrain, surface, and atmospheric variables. Each pixel represents the model's estimate of average solar energy received at that location, based on physical features rather than historical solar plant locations.

This map looks smooth and continuous because solar radiation is a large-scale climatic variable. It does not change abruptly over short distances.

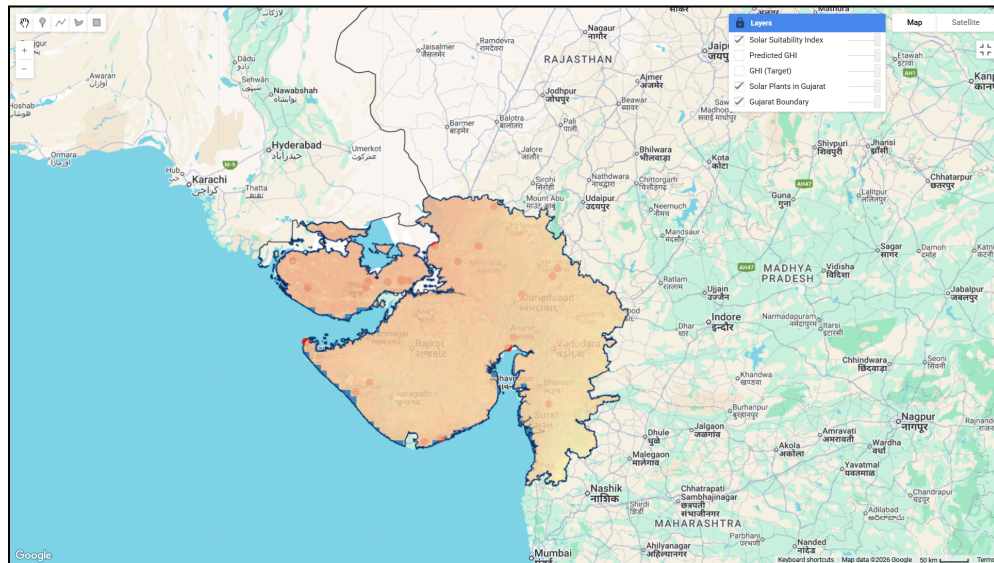
Higher predicted GHI in the western and central Gujarat (Saurashtra and Kutch regions). These regions benefit from flatter terrain, lower humidity and fewer persistent clouds. Whereas slightly lower predicted GHI in the coastal belts region.

This map is the foundation of the Solar Suitability index you generated

- Predicted GHI ---> physical solar potential
- Normalized GHI ---> relative suitability within Gujarat.

The absence of abrupt spatial variability indicates stable model behaviour and physical consistency.

(III) Solar suitability index map



This is the *normalized Solar Suitability Index* derived from the **predicted mean GHI** using min–max scaling.

So each pixel shows:

$$\text{Suitability} = \frac{\text{Predicted GHI} - GHI_{\min}}{GHI_{\max} - GHI_{\min}}$$

- Value $\approx 0 \rightarrow$ relatively lower solar potential
- Value $\approx 1 \rightarrow$ relatively higher solar potential

The map looks of almost one colour because Gujarat is uniformly solar-rich so GHI does not vary dramatically across Gujarat. Also min–max normalisation stretches values only between the local minimum and maximum. Normalized values cluster around mid-high range. This compresses visual contrast. We used Annual mean GHI that smooths out variability.

Even though the map looks uniform , what we can clearly interpret is that the western and interior regions of Gujarat (Kutchh and Saurashtra regions) show slightly higher suitability. These are the areas with flatter terrain and lower cloud attenuation. Whereas the coastal belts showed slightly lower suitability. These are the regions influenced by marine clouds and humidity.

Although the suitability surface is smooth, real-world installations (red dots) preferentially occur in the higher end of the predicted suitability range. This is **qualitative validation**.

8. Validation

Model validation was performed qualitatively by overlaying existing solar plant locations on the suitability map. Most installations fall within medium to high suitability zones, indicating consistency between model predictions and real-world deployment patterns. Since the target variable is derived from reanalysis data, numerical accuracy metrics against the same source would be circular; therefore, physical consistency and external deployment-based validation were emphasized.

9. Results

The results demonstrate that a regression-based, physics-informed approach can effectively capture spatial patterns of solar energy availability. The framework avoids bias toward historical installations and provides a transferable methodology for solar site assessment in other regions.

10. Conclusion

This project successfully integrates remote sensing, geospatial analysis, and machine learning to generate a continuous solar suitability map. By predicting GHI and deriving a suitability index, the study provides a physically meaningful decision-support tool for solar energy planning. The methodology is scalable and can be extended to other regions or renewable energy sources.
